DOMAIN GUIDE

Base USDC Settlement And Funding

This domain owns what happens to USDC on Base. It gives one loan an authoritative contract record and fixed rules for funding, borrower acceptance, repayment, lender claim, and refund. Bitcoin collateral lives under a separate chain authority. A Base payout can expose a secret that makes a previously authorized Bitcoin transaction usable, but it cannot sign, broadcast, or confirm that transaction.

> This is the domain guide for Base USDC Settlement And Funding, one of the five domains of Bitcoin-backed USDC one-loan system. The architecture guide, a separate document, explains how the domains work together.

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A **funding vault** is a deployed immutable per-loan address for incoming principal; a **repayment vault** is the corresponding address for debt repayment. Transfers into either are provisional until the named main-contract action finalizes the required exact amount. **Payer-neutral** means that the sender can be any eligible address: sending tokens does not acquire the borrower's or lender's role, change a recipient, or gain a refund right. Fixed recipients come from the approved loan and contract state, not the identity of the payer.

Settlement And Lender Interaction

Base loan state and USDC settlement owns the contract's states, deadlines, calls, named events/errors, exact amounts, and fixed recipients. It registers the funding vault; successful borrower acceptance (the `acceptX` call) atomically pays principal to the fixed borrower recipient and deploys/registers the repayment vault. Failed or conflicting calls revert without a success event or unintended value movement. The contract also provides the exact permissionless fixed-effect finalization, sweep, expiry, and refund calls. Permissionless execution lets another caller pay gas; it does not give that caller control over amount, recipient, or meaning.

The Base transition contract, the funding vault contract, and the repayment vault contract retain the full public contracts.

Lender-controlled funding-vault interaction supplies the human and application boundary for the selected lender's explicit capital action. It presents the finalized deployed vault, chain, asset, loan, remaining principal, deadline, and fixed lender settlement recipient. The lender may use one or more ordinary transfers from a controlled wallet or treasury, including a recoverability-qualified Vault payment account. Partial deposits accumulate. This responsibility

coordinates provisional balance display and finalization while leaving signing to the participant and authoritative transitions to the contract and observer. It never pools capital, substitutes a lender, or automatically draws funds.

From Funding To Repayment

After qualified origination, the main contract opens the funding window. Exact-principal finalization produces a funded-but-unaccepted state with a bounded acceptance lifetime. The borrower must intentionally disclose its acceptance secret, `X`, through its Vault-controlled call; acceptance pays principal and leaves Bitcoin activation pending. If the borrower's prefund refund confirms before activation, the lender's principal is left unsecured. Only separate confirmed Bitcoin evidence can make the product's active-collateral projection eligible. If the borrower never accepts, fixed-recipient expiry/refund remains possible and prefunded Bitcoin has its own recovery path.

The origination flow, the activation flow, the preactivation unwind flow, and the main Base loan states preserve these distinctions.

For repayment, sufficient vault balance is necessary but is not itself valid repayment. Exact debt must be finalized no later than the repayment deadline, `D`. A bound lender can then claim by revealing its claim secret, `Y`, strictly before the claim cutoff, `C`, two days after the repayment deadline. The claim pays the fixed lender recipient once and exposes `Y`; the observation domain qualifies its finality, and Bitcoin operations can complete the fixed borrower recovery transaction. The borrower may make the one permitted refund-recipient switch before `C`; at `C` the current recipient freezes.

The repayment vault states, the refund and cutoff states, and the normal repayment flow own the exact conditions.

If claim does not happen before `C`, the lender permanently loses that Base claim and the exact unclaimed repayment becomes permissionlessly refundable once to the frozen borrower recipient. The finalized `UnclaimedRepaymentRefunded` event is the only positive Base fallback trigger. Lender silence, a missing secret, elapsed time, or raw token activity is not that event. Base refund does not imply Bitcoin recovery, and Bitcoin lender-default maturity after `T`, three days after the repayment deadline remains independent. The valid repayment, claim and refund exclusivity, and default timing and Base separation invariants keep these consequences separate.

Failure, Authority, And Scope

Wrong loan, hash, role, recipient, amount, deadline, state, or deployment blocks the action. Excess and late configured USDC follow the specified fixed recovery paths; there is no generic asset rescue or arbitrary destination. Application retries must reload authoritative facts and preserve the same fixed effect. A workforce operator can request an already authorized operation but cannot edit contract meaning or claim success without evidence.

The recorded construction choices use immutable ownerless vaults, while the main contract retains its explicit Transparent Proxy administration, compatibility checks, and opener-replacement boundary. Who holds that administration remains unresolved (see Open questions).

Whether this version has an on-chain pause also remains open. Direct valid exits remain available during application or opener shutdown. These choices remain governed by the deployment, upgrade, and opener states, the funding opener and pause boundaries, and the Base upgrade boundary.

A Vault payment account and the separately scoped participant-transfer capability do not change payer-neutral contracts or historical recipients.

Cooperative Bitcoin settlement does not yet provide a complete Base accounting capability (cooperative settlement operations and Base accounting); liquidation, organization custody, and automatic lender-capital allocation remain outside this domain.

Open questions

Who holds that administration is an unresolved source conflict: the Base upgrade boundary states that V1 retains it under a 2-of-3 Safe through a seven-day timelock, while the upgrade governance selection states a dedicated single upgrade key for local/test use and the intentionally short public mainnet beta, followed by transfer of that sole ownership to a 2-of-3 multisig-controlled timelock. Neither is chosen until the conflict is answered.

Whether V1 has an on-chain pause is also open: this domain's settlement responsibility and the upgrade governance selection say V1 has no on-chain pause, while the deployment, upgrade, and opener states include pause roles and a narrow operational pause; whether that pause is the frontend/API operational stop or an on-chain role is undecided.

The architecture guide shows where this domain fits in the system, and the domain record lists exactly what the domain owns and excludes.